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{
  "id": "SOP-IBM3090-001",
  "title": "IBM 3090 Processor Complex: System Recovery & Availability Procedure",
  "system": "IBM 3090",
  "component": "3090 Processor Unit / TCM (Thermal Conduction Module)",
  "author": "SymbioStreams Semantic Bridge",
  "source": "IBM ES/3090 Complex Systems Recovery (GG24-3346)",
  "description": "High-level recovery procedures for system damage checks and TCM failure
isolation on the IBM 3090 platform.",
  "steps": [
    {
      "step": 1,
      "action": "Machine Check Interruption Analysis",
      "details": "When a machine check occurs, the hardware automatically stores logout
information. Examine the MCIC (Machine Check Interruption Code).",
      "indicator": "Check for bit 1 (System Damage) and bit 2 (Instruction Processing
Damage).",
    },
    {
      "step": 2,
      "action": "Analyze Log-Out Data",
      "details": "Use the Service Processor to display the 3090 internal log. Look for error
records associated with specific TCM (Thermal Conduction Module) addresses.",
      "check": "Locate the module ID (e.g., A-J1 or B-K2) that reported the check bit."
    },
    {
      "step": 3,
      "action": "Run Concurrent Maintenance Diagnostics",
      "details": "If the system is still operational (Soft Machine Check), initiate
concurrent diagnostics on the failing TCM.",
      "logic": "The Service Processor can isolate failures to a single MST-80 logic chip
within the TCM while the rest of the processor complex continues to run."
    },
    {
      "step": 4,
      "action": "System Recovery (Retry/Halt)",
      "details": "The 3090 has extensive instruction retry hardware. If retry fails, the
processor will enter a 'Hard Stop' state.",
      "remedy": "Physically replace the failing TCM and reload the microprogram from the
system console file."
    }
  ],
  "diagnostic_flowchart": [
    {
      "condition": "Bit 1 (System Damage) ON",
      "next": "Hard failure. Halt system and replace TCM."
    },
    {
      "condition": "Intermittent Instruction Damage",
      "next": "Retry logic active. Monitor error log for TCM degradation."
    },
    {
      "condition": "TCM Cooling Failure",
      "next": "Check coolant flow and temperature sensors on the frame."
    }
  ]
}

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